



Workshop Manual

Crafter 2017 ➤
TGE 2017 ➤

Fuel supply system - Diesel engines									
Engine ID	DAT A	DAU A	DAV A	DAS A	DAS B	DAW A	DAW B	DAU B	

Edition 03.2018



List of Workshop Manual Repair Groups

Repair Group

00 - Technical data

20 - Fuel supply system

Technical information should always be available to the foremen and mechanics, because their careful and constant adherence to the instructions is essential to ensure vehicle road-worthiness and safety. In addition, the normal basic safety precautions for working on motor vehicles must, as a matter of course, be observed.



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00 – Technical data

1 Safety information

(VRL011383; Edition 03.2018)

⇒ [“1.1 Safety regulations for working on fuel supply”, page 1](#)

⇒ [“1.2 Safety measures when working on vehicles with start/stop system”, page 2](#)

⇒ [“1.3 Safety precautions when using testers and measuring instruments during a road test”, page 2](#)

⇒ [“1.4 Safety precautions when working on cooling system”, page 2](#)

⇒ [“1.5 Safety precautions when working on the SCR system”, page 3](#)

⇒ [“1.6 General safety provisions”, page 3](#)

⇒ [“1.7 Safety precautions when working on exhaust system”, page 3](#)

1.1 Safety regulations for working on fuel supply

Risk of injury from highly pressurised fuel.

The fuel system is pressurised. Injury from fuel spray possible.

Before opening the fuel system:

- Wear protective goggles.
- Wear protective gloves.
- To release pressure, wrap a clean cloth around the connection and carefully loosen the connection.

Danger of fire caused by escaping fuel

When the battery is connected and the driver door opens, the door contact switch activates the fuel pump. Escaping fuel can ignite and cause a fire.

- Disconnect voltage supply to fuel pump before opening the fuel system.

Observe the fill level of the fuel tank

- ◆ Fuel tank must be only partly filled. To find out how much fuel may remain in the fuel tank, refer to the corresponding work procedure.
- ◆ If necessary, drain fuel tank ⇒ [page 16](#) .
- ◆ Before starting work, the extraction hose of a switched-on exhaust gas extraction system must be laid near to the assembly opening of the fuel tank.
- ◆ The exhaust extraction system is used to extract fuel vapours.



- ◆ If no exhaust gas extraction system is available, a radial fan with a displacement greater than 15 m³/h can be used providing that motor is not in air flow.

To prevent injuries to persons and destruction of the injection and glow plug system, the following must be noted:

- ◆ Always switch off the ignition before connecting or disconnecting electrical wiring of injection and glow plug system or connecting/disconnecting tester cables.
- ◆ Wash the engine only with the ignition switched off.
- ◆ When electrical connectors were separated and the engine was started, read event memory of engine control unit using
⇒ Vehicle diagnostic tester.

1.2 Safety measures when working on vehicles with start/stop system

Risk of injury from engine starting unexpectedly

If the vehicle's start/stop system is activated, the engine could start unexpectedly. A message in the dash panel insert indicates whether the start/stop system is activated.

- Deactivate start/stop system by switching off the ignition.

1.3 Safety precautions when using testers and measuring instruments during a road test

Risk of injury caused by unsecured testing and measuring instruments

When the front passenger airbag is triggered in an accident, insufficiently secured testing and measuring instruments become dangerous projectiles.

- Secure testing and measuring instruments on the rear seat.

Or

- Have a second person operate the test and measuring equipment on the rear seat.

1.4 Safety precautions when working on cooling system

Scalding from hot coolant

On a warm engine, the cooling system is under high pressure. Hot steam/hot coolant can escape - risk of scalding.

- Wear protective gloves.
- Wear protective goggles.
- Reduce excess pressure by covering the cap of the coolant expansion tank with cloths and opening carefully.



1.5 Safety precautions when working on the SCR system

Risk of injury due to reducing agent

The reducing agent may cause eye and skin irritations as well as injuries to the respiratory system and poisoning.

- Wear protective goggles.
- Wear protective gloves.
- Always wear protective clothes.
- Make sure that a sufficient amount of fresh air is supplied. In closed areas, switch on the exhaust gas extractor system.

Risk of damage caused by reducing agent

If reducing agent gets onto trim panels and body components, the reducing agent may solidify after a while and thus damage the surface.

- Make sure that no reducing agent gets onto the trim panels or body components.
- If any reducing agent does get onto surfaces, remove it immediately using water and a cloth.

1.6 General safety provisions

Risk of damage to components

Connecting or disconnecting electric cables or washing the engine while it is running may damage components.

- Switch off the ignition before connecting or disconnecting electric cables.
- Switch off the ignition before washing the engine.

Malfunction caused by air in the fuel supply system

After repair work has been performed on the fuel tank or the fuel delivery unit, air will be drawn in by the metering pump and conveyed to the auxiliary/supplementary heater. Air in the fuel supply system may cause the malfunction of the auxiliary/supplementary heater.

- Fill fuel take-off pipe with fuel.

1.7 Safety precautions when working on exhaust system

Risk of poisoning due to chemical substances

Exhaust gas temperature senders may contain chemical substances. There is a risk of poisoning or injuries to respiratory system.



- Never open an exhaust gas temperature sender by cutting, sawing or any other means.

Risk of injury due to hot condensate and particles in the exhaust system.

The exhaust system could contain hot condensate and/or particles. There is a risk of injury to the eyes, skin and respiratory system, as well as poisoning.

- Always wear protective gloves and eye protection when cutting the exhaust system.
- When cutting, use an extraction system or otherwise ensure sufficient ventilation.



2 Identification

⇒ **"2.1 Engine number/engine data", page 5**

2.1 Engine number/engine data

The engine number (engine code and serial number) is located at the front on the joint -arrow- between engine and gearbox.

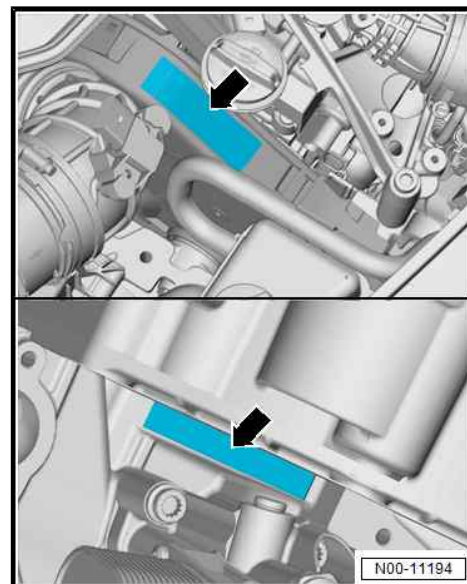
In addition, a sticker is attached to the top of the toothed belt guard -arrow- showing the engine code.

The engine codes have four characters.

The first 3 characters stand for the engine's cubic capacity and mechanical structure. They are stamped into the cylinder block together with the serial number.

The fourth character denotes the engine's performance and torque rating and depends on the engine control unit.

Fitting locations of vehicle data sticker ⇒ Maintenance ; Booklet 10.3 .



Codes	DAUB	DAUA	DAVA
Emissions standard	EU 3, 4, 5 and 6	EU 3, 4, 5 and 6	EU 6
Displacement l	1.968	1.968	1.968
Power kW at rpm	75/ 3500	103/ 3500	130/ 3600
Torque Nm at rpm	300/ 2000	340/ 2000	410/ 2000
Bore Diameter, mm	81.0	81.0	81.0
Stroke mm	95.5	95.5	95.5
Compression ratio	15.5	15.5	15.5
RON	Diesel EN 590	Diesel EN 590	Diesel EN 590
Exhaust turbocharger	Mono-turbo	Mono-turbo	Bi-turbo
Firing order	1-3-4-2	1-3-4-2	1-3-4-2
Cylinder	4	4	4



3 Repair notes

⇒ [“3.1 Rules for cleanliness”, page 6](#)

⇒ [“3.2 General information”, page 6](#)

⇒ [“3.3 General repair instructions”, page 7](#)

⇒ [“3.4 Foreign objects in engine”, page 7](#)

⇒ [“3.5 Contact corrosion”, page 7](#)

⇒ [“3.6 Routing and attachment of lines”, page 8](#)

⇒ [“3.7 Fitting radiator and condensers”, page 8](#)

⇒ [“3.8 Checking vacuum system”, page 8](#)

3.1 Rules for cleanliness

Even slight soiling can cause faults. You must therefore comply with the following cleanliness rules when working on the fuel supply system, injectors or the turbocharger:

- ◆ Carefully clean connection points and the surrounding area, and dry thoroughly before opening.
- ◆ Seal open pipes and connections immediately with clean plugs for example from the engine bung set - VAS 6122- .
- ◆ Place removed parts on a clean surface and cover them over. Use only lint-free cloths.
- ◆ If repair work cannot be performed immediately, carefully cover or seal components.
- ◆ Install clean components only: do not remove replacement parts from packing until immediately before installing. Do not use parts that have been stored outside of their packaging (e.g. in tool boxes).
- ◆ If system is open, do not work with compressed air. Try not to move the vehicle.
- ◆ Protect disconnected electrical connectors from dirt and water, and reconnect them only when dry.
- ◆ The components must be clean and dry.

3.2 General information

- ◆ The engine control unit has a self-diagnosis capability. Before carrying out repairs and for fault finding, first read event memory. Also the vacuum hoses and connections must be checked (unmetered air).
- ◆ For trouble-free operation of electrical components, a voltage of at least 11.5 volts is necessary.
- ◆ Do not use sealants containing silicone. Particles of silicone drawn into the engine will not be burnt in the engine and damage the Lambda probe.
- ◆ Vehicles are fitted with a crash fuel shut-off circuit. It reduces the danger of a fire in a crash as the fuel pump is switched off via the fuel pump relay.
- ◆ The system also improves the starting characteristics of the engine. When the driver door is opened, the fuel pump is activated for 2 seconds in order to build up pressure in the fuel system, observe safety precautions ⇒ [page 1](#) .



3.3 General repair instructions

- ◆ Clean tools and workbench etc. before working on injection system.
- ◆ The injectors and locations are to be checked before installation by means of a visual inspection for damage and dirt. It must be ensured that the injector bore is clean. If necessary, clean the injector shaft using a clean cloth. Take care not to cause any damage (do not use any sharp tools!).
- ◆ If high-pressure lines are not renewed, they must be labelled on removal. High-pressure fuel lines must always be re-installed in their original positions (i.e. on the same cylinder).
- ◆ Take care not to damage the injectors when removing the old copper seals.
- ◆ Check all new O-rings for damage before installing. Lubricate O-rings with engine oil or assembly oil before installing.
- ◆ Align high-pressure lines so that they are not subjected to stress. First tighten all unions hand-tight and then tighten to torque.
- ◆ Never attempt to reshape high-pressure fuel lines.
- ◆ When working on any parts of the high-pressure fuel system, tools may only be used for loosening and tightening pipe unions. All other components must always be removed and installed by hand without using tools or other equipment.
- ◆ Press the fuel return lines onto the injectors by hand from above so that they engage audibly on each injector (do not press in the release pins when doing this). Then, press the release pin downwards. Pull fuel return lines upwards by hand to check that they are firmly attached and are not leaking.
- ◆ Disassembling individual common rail components is not permitted. The components should be replaced as complete units if faulty.
- ◆ Do not carry out any installation work on common rail system whilst engine is running.
- ◆ Never attempt to bleed the common rail system by loosening high-pressure components after starting the engine.
- ◆ All cable ties which were opened or cut during removal must be renewed at the same locations.
- ◆ Fuel hoses in engine compartment must be secured only with spring-type clips. The use of crimp-type or screw-type clips is not permissible.

3.4 Foreign objects in engine

- ◆ To prevent the ingress of foreign bodies during work on the engine, seal all openings.
- ◆ Seal openings with suitable bungs from engine bung set - VAS 6122-.

3.5 Contact corrosion

Contact corrosion can occur if non-approved fasteners are used on the vehicle (bolts, nuts, washers etc.).

For this reason, only connecting elements with a special surface coating have been fitted.

In addition, rubber, plastic and adhesives are made of non-conductive materials.



If there is any doubt about the suitability of parts, a general rule is to use new parts → Electronic Parts Catalogue (ETKA) .

Please note

- ◆ Use Genuine parts, because they have been tested and compatible with aluminium.
- ◆ Use Volkswagen parts.
- ◆ Damage resulting from contact corrosion is not covered by the warranty.

3.6 Routing and attachment of lines

Risk of damage to lines

Lines may become damaged by moving or hot components.

- Route lines in their original positions.
- Ensure there is sufficient clearance to moving or hot components.

3.7 Fitting radiator and condensers

Even if installed correctly, the radiator, the condenser and the charge air cooler may have small dents in their fins. This does not mean that these components have been damaged. It is not permissible to renew radiators, charge air coolers or condensers only because of such minor dents.

3.8 Checking vacuum system

Special tools and workshop equipment required

- ◆ Hand operated vacuum pump - VAS 6213-



Procedure

- Check all vacuum lines throughout entire vacuum system for:
 - ◆ Cracks
 - ◆ Marten bites
 - ◆ Crushing
 - ◆ Porous areas and other leaks
- Check vacuum line to solenoid valve and from solenoid valve to relevant component.
- If a fault has been entered in the event memory, make sure to check not only all vacuum lines leading to the specified component, but also those leading to other components.



- If pressure cannot be built-up using hand vacuum pump - VAS 6213- or pressure immediately drops again, check hand vacuum pump and connecting hoses for leaks.



20 – Fuel supply system

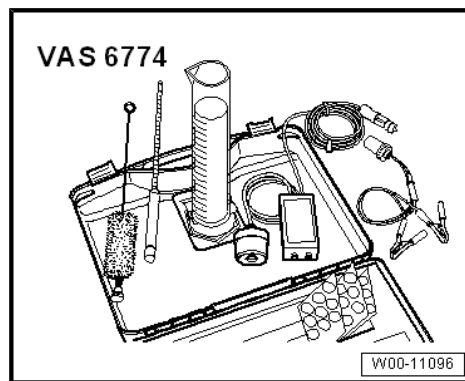
1 Procedure should vehicle be misfueled

⇒ [“1.1 Procedure should vehicle be misfueled”, page 10](#)

1.1 Procedure should vehicle be misfueled

Special tools and workshop equipment required

- ◆ Fuel identification unit - VAS 6774-



Note

- ◆ *Misfuelling can cause irreversible damage to the high-pressure components, particularly the high-pressure pump, due to the lack of lubrication otherwise provided by diesel fuel.*
- ◆ *Damage due to seizure and metal particles can be expected.*
- ◆ *These particles can also cause further damage in the pressure control valve and injectors.*
- ◆ *If in any doubt regarding the fuel quality, take a sample and analyse the fuel with the fuel analysis unit - VAS 6774- ⇒ operating instructions .*
- ◆ *The fuel analysis unit - VAS 6774- is no substitute for a laboratory test in warranty cases.*

These instructions are intended as a guide through the individual work steps by means of a flowchart, while bearing various conditions in mind.

Was the engine started with the incorrect fuel?

Yes

⇒ [“1.1.1 Step 1, engine was started with incorrect fuel”, page 11](#) .

No

⇒ [“1.1.2 Step 2, engine was not started with incorrect fuel”, page 11](#) .



⇒ [“1.1.1 Step 1, engine was started with incorrect fuel”, page 11](#)

⇒ [“1.1.2 Step 2, engine was not started with incorrect fuel”, page 11](#)

⇒ [“1.1.3 Step 3, swarf in fuel delivery unit GX1 and fuel tank”, page 12](#)

⇒ [“1.1.4 Step 4, no swarf in fuel delivery unit GX1 and fuel tank”, page 13](#)

⇒ [“1.1.5 Step 5, there are metal particles in high-pressure pump”, page 13](#)

⇒ [“1.1.6 Step 6, there are no metal particles in high-pressure pump”, page 13](#)

1.1.1 Step 1, engine was started with incorrect fuel

Procedure

- If fitted, separate electrical connector of supplementary fuel pump - V393- in engine compartment beneath coolant expansion tank.
- Remove fuel tank ⇒ [page 19](#) .
- Remove fuel delivery unit - GX1- , and drain ⇒ [page 27](#) .
- Drain fuel pump reservoir of fuel delivery unit - GX1- .
- Drain fuel tank ⇒ [page 16](#) .
- Check fuel tank for metal particles.
- Carry out a visual inspection for foreign bodies and metal particles in fuel pump reservoir and on preliminary filter.

Metal particles found?

Yes

⇒ [“1.1.3 Step 3, swarf in fuel delivery unit GX1 and fuel tank”, page 12](#) .

No

⇒ [“1.1.4 Step 4, no swarf in fuel delivery unit GX1 and fuel tank”, page 13](#) .

1.1.2 Step 2, engine was not started with incorrect fuel

Procedure

- Remove fuel tank ⇒ [page 19](#) .
- Remove fuel delivery unit - GX1- , and drain ⇒ [page 27](#) .
- Check strainer and clean if soiled.
- Drain fuel tank ⇒ [page 16](#) .

Fuel tank completely empty:

- Clean inside of fuel tank with lint-free cloth.
- Install fuel delivery unit - GX1- ⇒ [page 27](#) .
- Install fuel tank ⇒ [page 19](#) .
- Fill fuel tank with 5 litres of diesel fuel.
- Perform bleed fuel system function using ⇒ Vehicle diagnostic tester.



Note

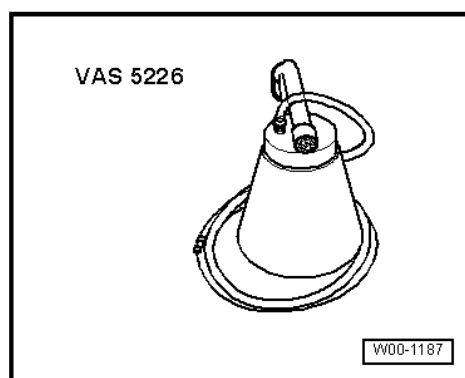
This process takes 130 seconds. Fuel pumps are actuated a total of three times in the process. The process must not be terminated prematurely.

- Renew fuel filter ⇒ [page 34](#) .
- Fill up vehicle.
- Carry out road test.

1.1.3 Step 3, swarf in fuel delivery unit - GX1- and fuel tank

Special tools and workshop equipment required

- ◆ Diesel extractor - VAS 5226-



Procedure

- Clean fuel delivery unit - GX1- and fuel tank using diesel extractor - VAS 5226- .
- Install fuel delivery unit - GX1- ⇒ [page 27](#) .
- Fill fuel tank with 5 litres of diesel fuel.
- Completely empty fuel tank again as described above ⇒ [page 11](#) .

Renew following high-pressure components:

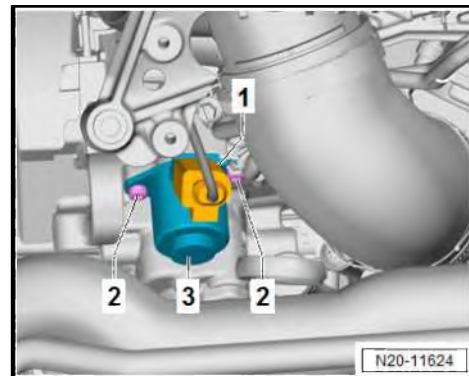
- ◆ High-pressure pump ⇒ Rep. gr. 23 ; High-pressure pump; Removing and installing high-pressure pump .
- ◆ High-pressure lines ⇒ Rep. gr. 23 ; Injectors/fuel rail; Removing and installing high-pressure lines .
- ◆ Fuel rail including fuel pressure regulating valve - N276- and fuel pressure sender - G247- ⇒ Rep. gr. 23 ; Injectors/fuel rail; Removing and installing fuel rail
- ◆ Injectors ⇒ Rep. gr. 23 ; Injectors/fuel rail (fuel rail); Removing and installing injectors
- ◆ Fuel return lines (overflow oil lines) ⇒ Rep. gr. 23 ; Injector units/fuel rail; Assembly overview - injectors .
- ◆ Fuel filter: Renew ⇒ [page 34](#)
- Fill up vehicle.
- Bleed fuel system ⇒ Rep. gr. 23 ; Injection system; Filling/bleeding fuel system .
- Carry out road test.



1.1.4 Step 4, no swarf in fuel delivery unit - GX1- and fuel tank

Procedure

- Fill fuel tank with 5 litres of diesel fuel.
- Completely empty fuel tank again as described above
⇒ [page 11](#) .
- Separate electrical connector -1-.
- Unscrew bolts -2-.
- Remove fuel metering valve - N290- -3-.
- Check fuel metering valve - N290- -3- and high-pressure pump for metal particles.



Metal particles found?

Yes

⇒ [“1.1.5 Step 5, there are metal particles in high-pressure pump”, page 13](#) .

No

⇒ [“1.1.6 Step 6, there are no metal particles in high-pressure pump”, page 13](#) .

1.1.5 Step 5, there are metal particles in high-pressure pump

Procedure

- Replace high-pressure pump ⇒ Rep. gr. 23 ; High-pressure pump: Removing and installing high-pressure pump .
- Renew high-pressure lines ⇒ Rep. gr. 23 ; Injectors/fuel rail (fuel rail): Removing and installing high-pressure lines .
- Renew fuel rail including fuel pressure regulating valve - N276- and fuel pressure sender - G247- ⇒ Rep. gr. 23 ; Injectors/ fuel rail: Removing and installing fuel rail .
- Renew injectors ⇒ Rep. gr. 23 ; Injectors/fuel rail; Removing and installing injectors
- Renew fuel return lines ⇒ Rep. gr. 23 ; Injector units/fuel rail; Assembly overview - fuel rail
- Renew fuel filter ⇒ [page 34](#) .
- Fill up vehicle.
- Fill/bleed fuel system ⇒ Rep. gr. 23 ; Injection system; Filling/ bleeding fuel system .
- Carry out road test.

1.1.6 Step 6, there are no metal particles in high-pressure pump

Procedure

- Replace high-pressure pump ⇒ Rep. gr. 23 ; High-pressure pump: Removing and installing high-pressure pump .
- Renew fuel filter ⇒ [page 34](#) .
- Fill/bleed fuel system ⇒ Rep. gr. 23 ; Injection system; Filling/ bleeding fuel system .
- Fill up vehicle.



- Carry out road test.



2 Fuel

⇒ [“2.1 Assembly overview - fuel tank”, page 15](#)

⇒ [“2.2 Emptying fuel tank”, page 16](#)

⇒ [“2.3 Removing and installing fuel tank”, page 19](#)

2.1 Assembly overview - fuel tank

1 - Fuel

- ☐ Removing and installing
⇒ [page 19](#)

2 - Bolt

- ☐ 8 Nm +90°

3 - Clip

4 - Retaining strap

5 - Cap

- ☐ Renew seal if damaged.

6 - Fuel line

- ☐ From metering pump - V54- to auxiliary heater
- ☐ Do not kink
- ☐ Depending on equipment ⇒ Auxiliary heater, supplementary heater; Rep. gr. 82 ; Fuel supply system; Overview of fitting locations - fuel supply system .
- ☐ Separating plug-in connectors ⇒ [page 29](#)

7 - Fuel return line

- ☐ From engine
- ☐ Do not kink
- ☐ Disconnect plug-in connectors ⇒ [page 29](#)

8 - Fuel supply line

- ☐ To engine
- ☐ Do not kink
- ☐ Disconnect plug-in connectors ⇒ [page 29](#)

9 - Union nut

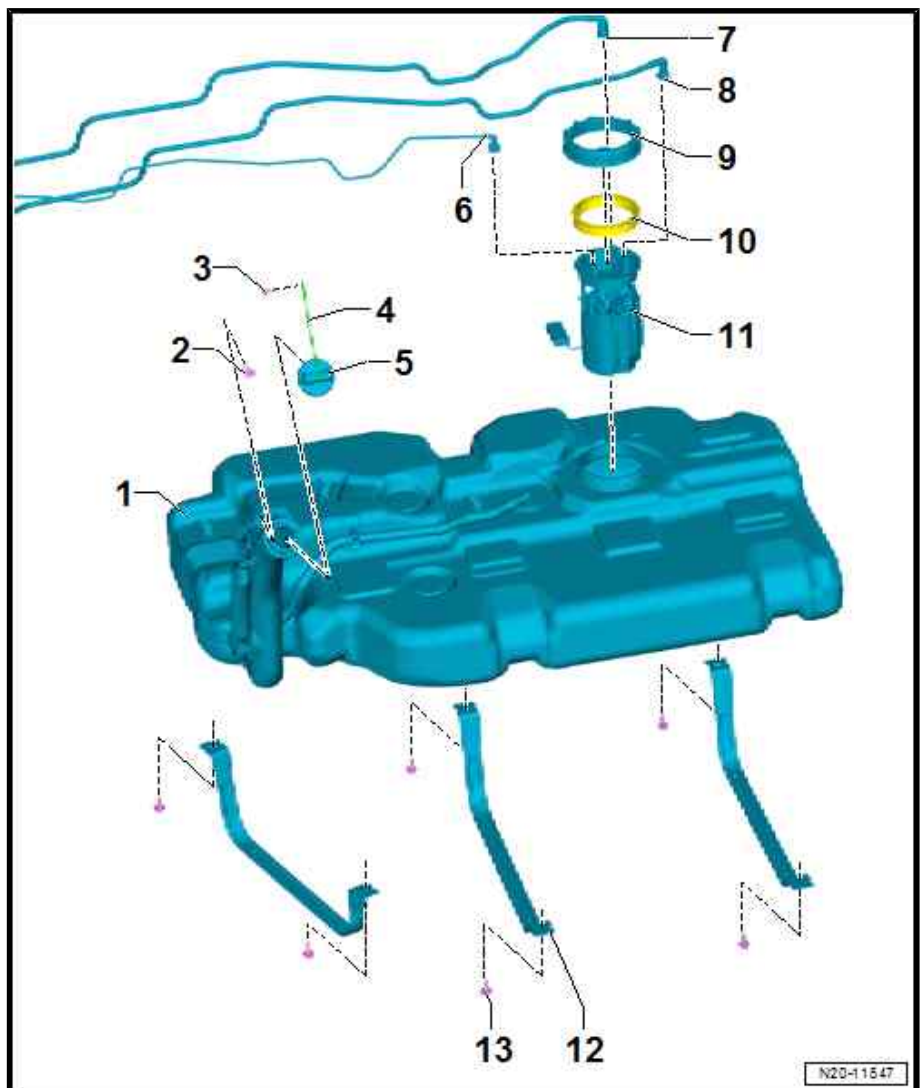
- ☐ Specified torque ⇒ [Item 7 \(page 26\)](#)

10 - Seal

- ☐ Renew after removal
- ☐ Insert dry.
- ☐ Moisten with fuel only when installing fuel delivery unit - GX1-

11 - Fuel delivery unit - GX1-

- ☐ With fuel system pressurisation pump - G6- .
- ☐ With integrated fuel filter; cannot be renewed individually
- ☐ With integrated pressure limiting valve; cannot be renewed individually





- ☐ Assembly overview ➔ [page 26](#)
- ☐ Removing and installing ➔ [page 27](#)

12 - Securing strap

- ☐ Observe installation position
- ☐ Ensure firm seating

13 - Bolt

- ☐ Renew after removal
- ☐ 20 Nm +90°

2.2 Emptying fuel tank

⇒ [“2.2.1 Emptying fuel tank when fuel pump is intact”, page 16](#)

⇒ [“2.2.2 Emptying fuel tank if it is more than 3/4 full:”, page 17](#)

⇒ [“2.2.3 Emptying fuel tank if it is less than 3/4 full”, page 19](#)

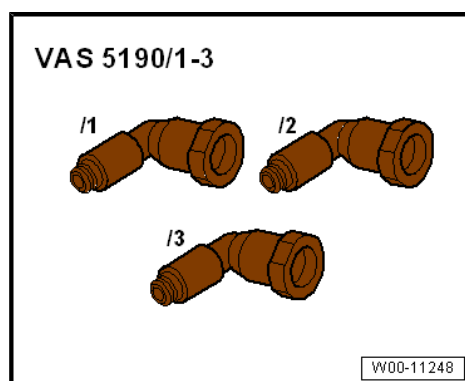
2.2.1 Emptying fuel tank when fuel pump is intact

Special tools and workshop equipment required

- ◆ Fuel extraction unit - VAS 5190-



- ◆ Adapter for fuel extraction - VAS 5190-





Procedure

Prepare special tool as follows



Note

If the fuel extraction unit - VAS 5190- is still fitted with the extraction hose with a fixed tip, the extraction hose must be replaced with the version with a screw-in tip.

- Unscrew tip -1- from extraction hose of fuel extractor - VAS 5190- .
- Screw fuel extraction adapter - VAS 5190 /2- -2- from fuel extraction adapter set - VAS 5190 /10- onto extraction hose.
- Switch off ignition.



CAUTION

The fuel system is pressurised.

Danger of injury caused by fuel spray.

- Wear protective goggles.
- Wear protective gloves.
- To release pressure, wrap a clean cloth around the connection and carefully loosen the connection.

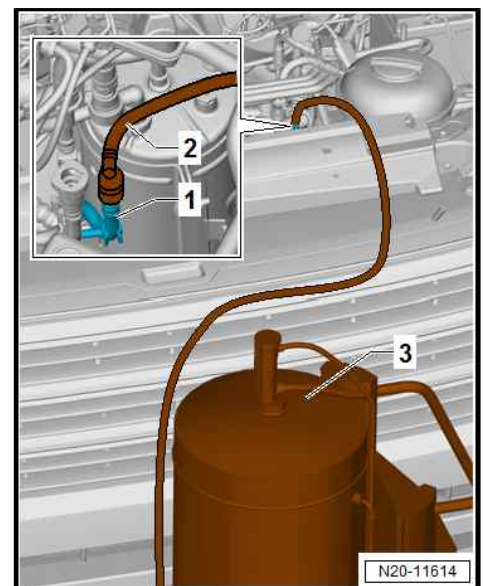
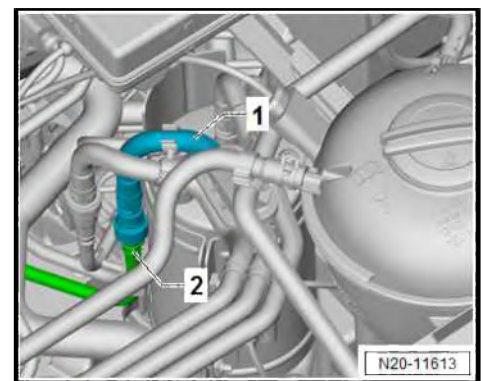
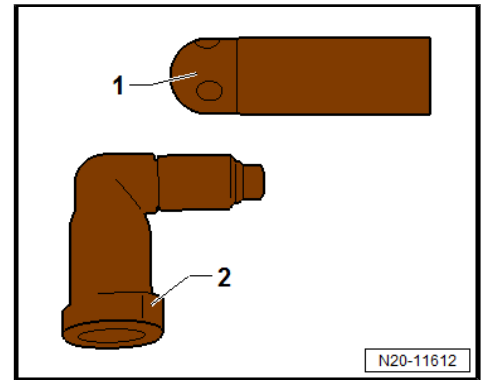
- Separate fuel return line -1- and fuel supply line -2-.
- Collect escaping fuel with a cleaning cloth.
- Connect extraction hose -2- of fuel extractor - VAS 5190- -3- with fuel supply line -1-.



Note

The fuel system pressurisation pump - G6- is actuated by ⇒ Vehicle diagnostic tester, and the fuel tank is emptied in the process.

- Connect ⇒ Vehicle diagnostic tester.
- Use ⇒ Vehicle diagnostic tester to drain fuel tank.
- Repeat the function if necessary.
- Open shut-off tap on fuel extractor - VAS 5190- until fuel tank is emptied.



2.2.2 Emptying fuel tank if it is more than $\frac{3}{4}$ full:

Special tools and workshop equipment required

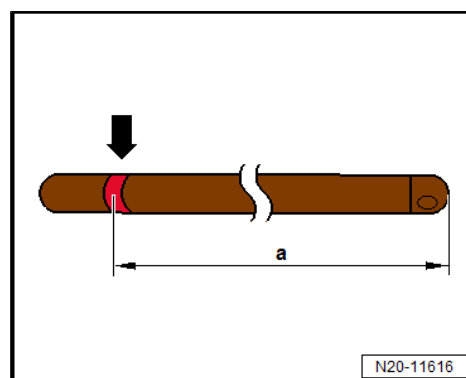


◆ Fuel extraction unit - VAS 5190 A-

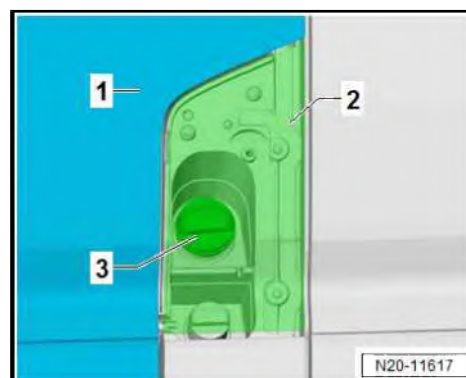


Procedure

- Make a mark -arrow- at dimension -a- from end of extraction hose of fuel extractor - VAS 5190- .
- Dimension -a- on extraction hose: 660 mm



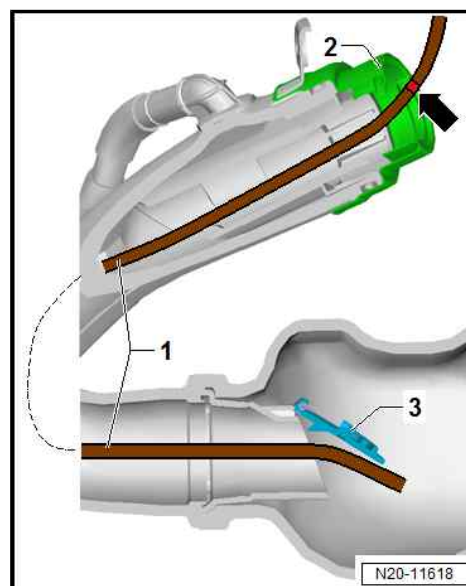
- Open driver door -1-.
- Open tank flap unit -2-.
- Clean area around fuel filler neck.
- Unscrew filler cap -3-.
- Secure earth wire of fuel extractor - VAS 5190- to a bare metal part of the body.



- Insert extraction hose -1- of fuel extractor - VAS 5190- into fuel tank until mark -arrow- aligns with filler neck -2-.
- Drain fuel tank via filler neck as far as possible.

Note

- ◆ *There is a non-return valve -3- in the fuel tank at the lower end of the filler neck. The valve must not be damaged by the extraction hose -1-. Therefore push extraction hose -1- into filler neck only as far as marking -arrow-.*
- ◆ *If the extraction hose -1- catches on the non-return valve -3- when being extracted, do not apply force to pull out the extraction hose -1-.*
- ◆ *In this case, remove the fuel delivery unit - GX1- and open the non-return valve -3- manually. Ensure that your arm does not come into contact with fuel.*
- Pull extraction hose out carefully.





2.2.3 Emptying fuel tank if it is less than $\frac{3}{4}$ full

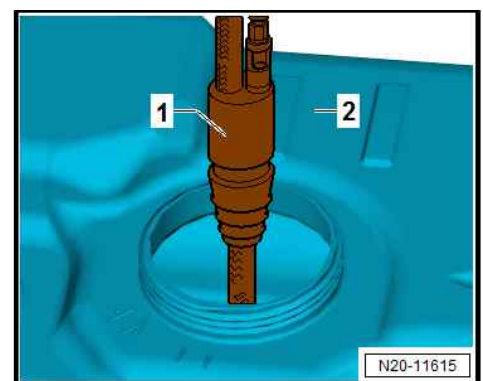
Special tools and workshop equipment required

- ◆ Fuel extraction unit - VAS 5190-



Procedure

- Remove fuel delivery unit - GX1- ⇒ [page 27](#) .
- Completely empty the fuel tank -2- using the extraction hose -1- of fuel extractor - VAS 5190- through the opening of the fuel delivery unit - GX1- .



2.3 Removing and installing fuel tank

⇒ [“2.3.1 Removing and installing fuel tank”, page 19](#)

⇒ [“2.3.2 Lower fuel tank.”, page 22](#)

2.3.1 Removing and installing fuel tank

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-

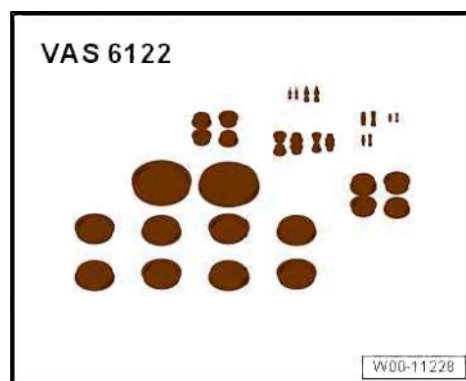




◆ Engine and gearbox jack - VAS 6931-



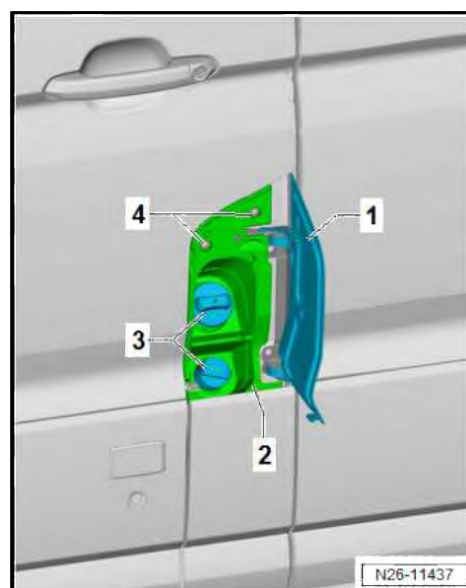
◆ Engine bung set - VAS 6122-



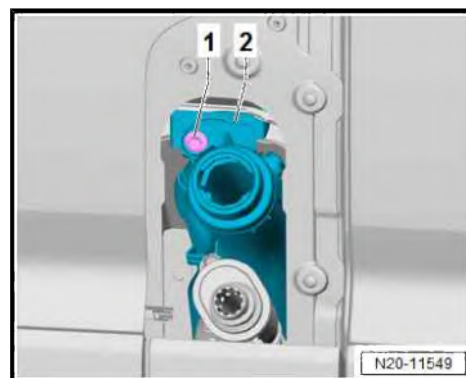
◆ Tensioning strap

Removing

- Disconnect battery ⇒ Electrical system; Rep. gr. 27 ; Battery; Disconnecting and connecting battery .
- Open driver door.
- Remove lower B-pillar cover ⇒ General body repairs, exterior; Rep. gr. 66 ; Mouldings/trim/ extensions; Removing and installing cover .
- Open tank flap -1-.
- Unscrew filler cap -3-.
- Unscrew bolts -4-.
- Remove filler neck bowl -2-.



- Unscrew bolt -1- on filler neck -2-.



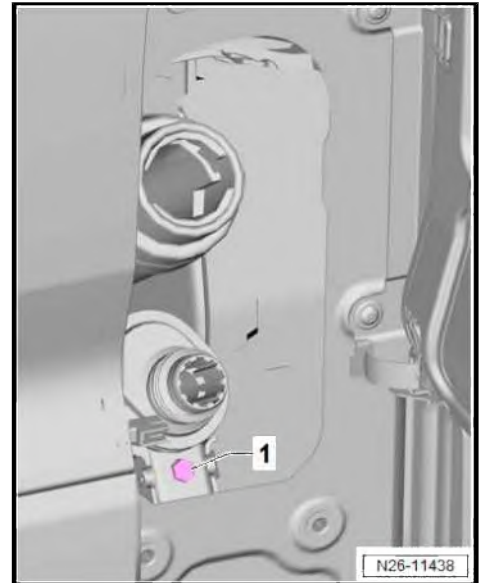


- Unscrew bolt -1-.
- Screw on filler cap again.
- Drain tank for reducing agent ⇒ Rep. gr. 26 ; SCR system (selective catalytic reduction); Draining tank for reducing agent .
- If fitted, remove underbody cover ⇒ General body repairs, exterior; Rep. gr. 66 ; Underbody cover; Removing and installing underbody cladding .

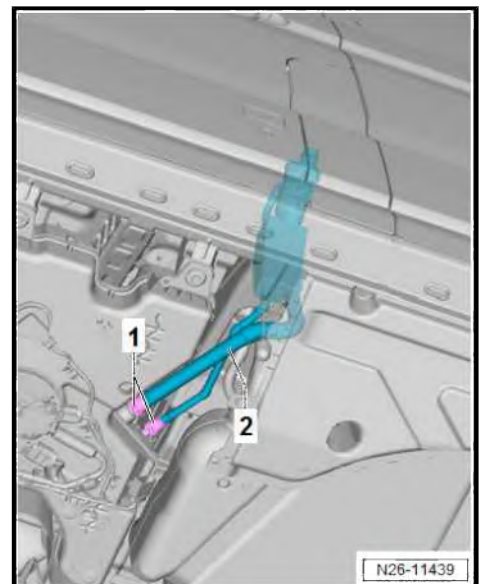
⚠ CAUTION

Risk of injury caused by escaping reducing agent.
 The reducing agent may cause eye and skin irritations as well as injuries to the respiratory system and poisoning.

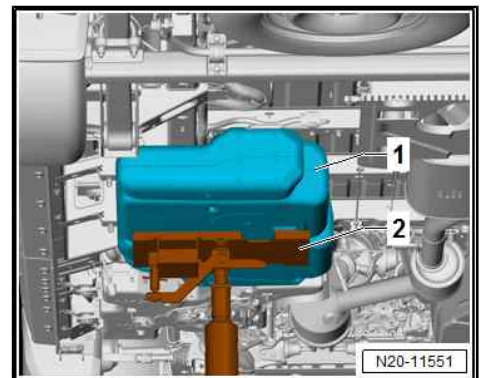
- Wear protective goggles.
- Wear protective gloves.
- Always wear protective clothes.
- Make sure that a sufficient amount of fresh air is supplied.
 In closed areas, switch on the exhaust gas extractor system.



- Disconnect quick-release coupling -1-.
- Remove filler pipe -2- upwards.
- Seal off all open lines and connections with clean plugs from engine bung set - VAS 6122- .
- Use tensioning straps to secure engine and gearbox jack - VAS 6931- from falling.



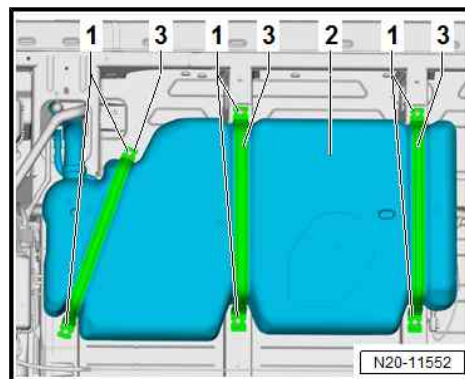
- Place engine and gearbox jack - VAS 6931- -2- under fuel tank -1-.





Note

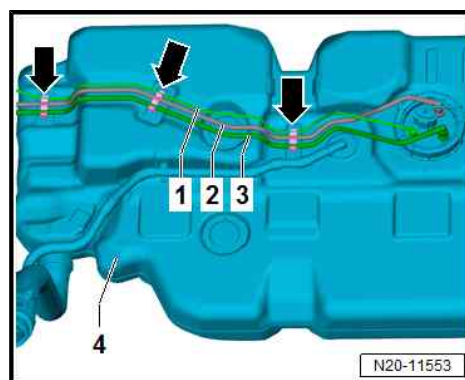
- ♦ For reasons of clarity, the fuel tank is shown without the engine and gearbox jack - VAS 6931- .
- ♦ The aid of a second mechanic is required to remove the fuel tank.
- Unscrew bolts -1-.
- Remove accessible tensioning straps -3- from fuel tank -2-.



Note

Carefully lower the fuel tank with engine and gearbox jack - VAS 6931- until the fuel lines above fuel tank are accessible.

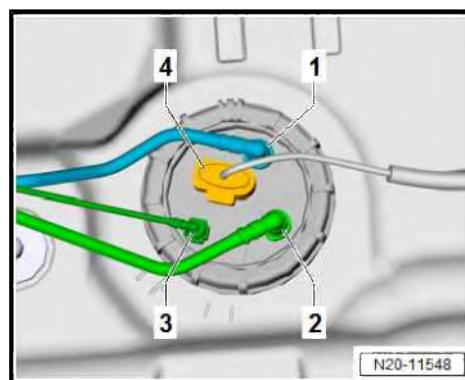
- If fitted, unclip fuel lines -1- from attachment -arrow-.
- If fitted, unclip fuel lines -2- and -3- from attachments -arrows-.



Note

Continue lowering fuel tank -4- with engine and gearbox jack - VAS 6931- until the connector and fuel lines on the fuel delivery unit - GX1- can be disconnected.

- If fitted, separate fuel line -3-.
- Disconnect fuel line -1-.
- Disconnect fuel line -2-.
- Separate electrical connector -4-.
- Lower fuel tank further with engine and gearbox jack - VAS 6931- .
- Loosen tensioning strap.
- Lift down the fuel tank from the engine and gearbox jack - VAS 6931- with the help of a second mechanic.
- Seal open fuel hoses and connections with clean plugs from engine bung set - VAS 6122- .



Installing

Install in reverse order of removal, observing the following:

Torque settings

- ♦ SCR system (Selective Catalytic Reduction); Assembly overview - reducing agent tank ⇒ Rep. gr. 26 ; SCR system (Selective Catalytic Reduction); Assembly overview - reducing agent tank
- ♦ ⇒ [“2.1 Assembly overview - fuel tank”, page 15](#)

2.3.2 Lower fuel tank.

Special tools and workshop equipment required



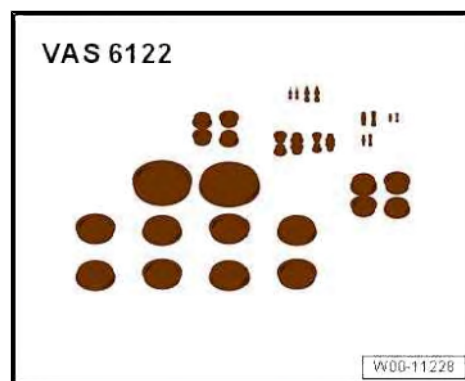
- ◆ Torque wrench - V.A.G 1331-



- ◆ Engine and gearbox jack - VAS 6931-



- ◆ Engine bung set - VAS 6122-

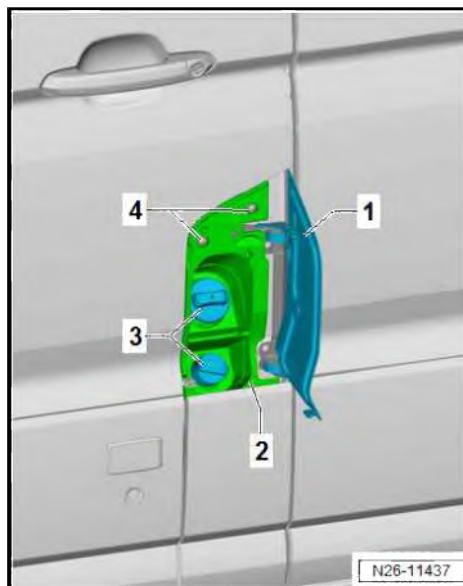


- ◆ Tensioning strap

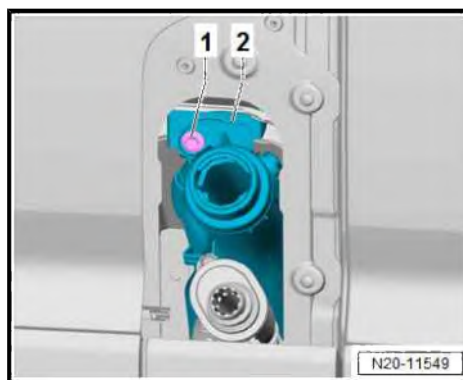


Procedure

- Disconnect battery ⇒ Electrical system; Rep. gr. 27 ; Battery; Disconnecting and connecting battery .
- Open driver door.
- Open tank flap -1-.
- Unscrew filler cap -3-.
- Unscrew bolts -4-.
- Remove filler neck bowl -2-.



- Unscrew bolt -1- on filler neck -2-.



- Unscrew bolt -1-.
- Screw on filler cap again.
- Drain tank for reducing agent ⇒ Rep. gr. 26 ; SCR system (selective catalytic reduction); Draining tank for reducing agent .
- If fitted, remove underbody cover ⇒ General body repairs, exterior; Rep. gr. 66 ; Underbody cover; Removing and installing underbody cladding .

⚠ CAUTION

Risk of injury caused by escaping reducing agent.

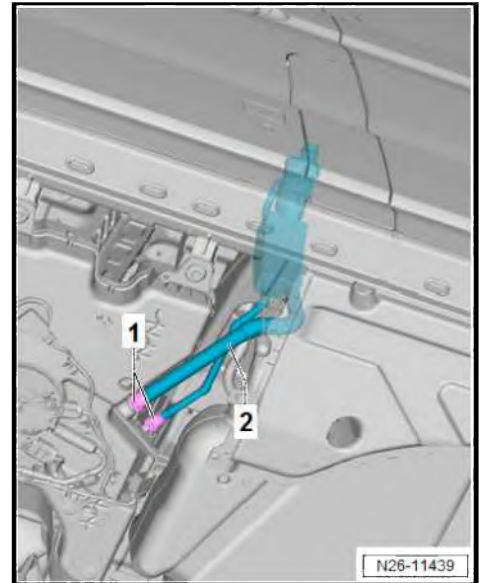
The reducing agent may cause eye and skin irritations as well as injuries to the respiratory system and poisoning.

- Wear protective goggles.
- Wear protective gloves.
- Always wear protective clothes.
- Make sure that a sufficient amount of fresh air is supplied. In closed areas, switch on the exhaust gas extractor system.

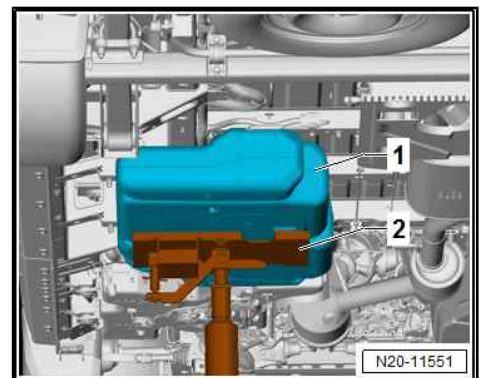




- Disconnect quick-release coupling -1-.
- Remove filler pipe -2- upwards.
- Seal off all open lines and connections with clean plugs from engine bung set - VAS 6122- .
- Use tensioning straps to secure engine and gearbox jack - VAS 6931- from falling.



- Place engine and gearbox jack - VAS 6931- -2- under fuel tank -1-.



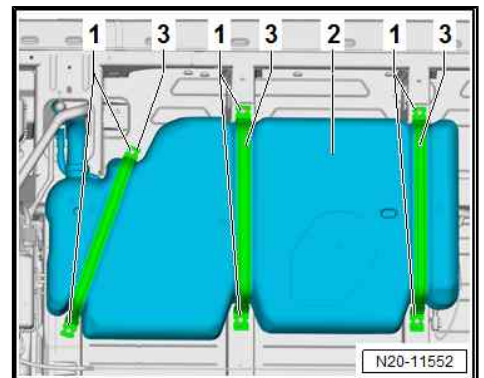
Note

For reasons of clarity, the fuel tank is shown without the engine and gearbox jack - VAS 6931- .

- Unscrew bolts -1-.
- Remove accessible tensioning straps -3- from fuel tank -2-.
- Carefully lower the fuel tank with engine and gearbox jack - VAS 6931- until the fuel lines above fuel tank are accessible.

Torque settings

- ◆ SCR system (Selective Catalytic Reduction); Assembly overview - reducing agent tank ⇒ Rep. gr. 26 ; SCR system (Selective Catalytic Reduction); Assembly overview - reducing agent tank
- ◆ ⇒ [“2.1 Assembly overview - fuel tank”, page 15](#)



3 Fuel delivery unit, fuel gauge sender

⇒ "3.1 Assembly overview - fuel delivery unit and fuel gauge sender", page 26

⇒ "3.2 Removing and installing fuel delivery unit, fuel gauge sender", page 27

⇒ "3.3 Removing and installing fuel gauge sender G ", page 28

3.1 Assembly overview - fuel delivery unit and fuel gauge sender

1 - Fuel return line

- ☐ Do not kink
- ☐ Disconnect plug-in connectors ⇒ [page 29](#)

2 - Fuel supply line

- ☐ Do not kink
- ☐ Disconnect plug-in connectors ⇒ [page 29](#)

3 - Fuel line

- ☐ From metering pump - V54- to auxiliary heater
- ☐ Do not kink
- ☐ Depending on equipment ⇒ Auxiliary heater, supplementary heater; Rep. gr. 82 ; Fuel supply system; Overview of fitting locations - fuel supply system .
- ☐ Disconnect plug-in connectors ⇒ [page 29](#)

4 - Fuel delivery unit - GX1-

- ☐ With fuel system pressurisation pump - G6- .
- ☐ With integrated fuel filter; cannot be renewed individually
- ☐ With integrated pressure limiting valve; cannot be renewed individually
- ☐ Removing and installing ⇒ [page 27](#)

5 - Fuel gauge sender - G-

- ☐ Removing and installing ⇒ [page 28](#)

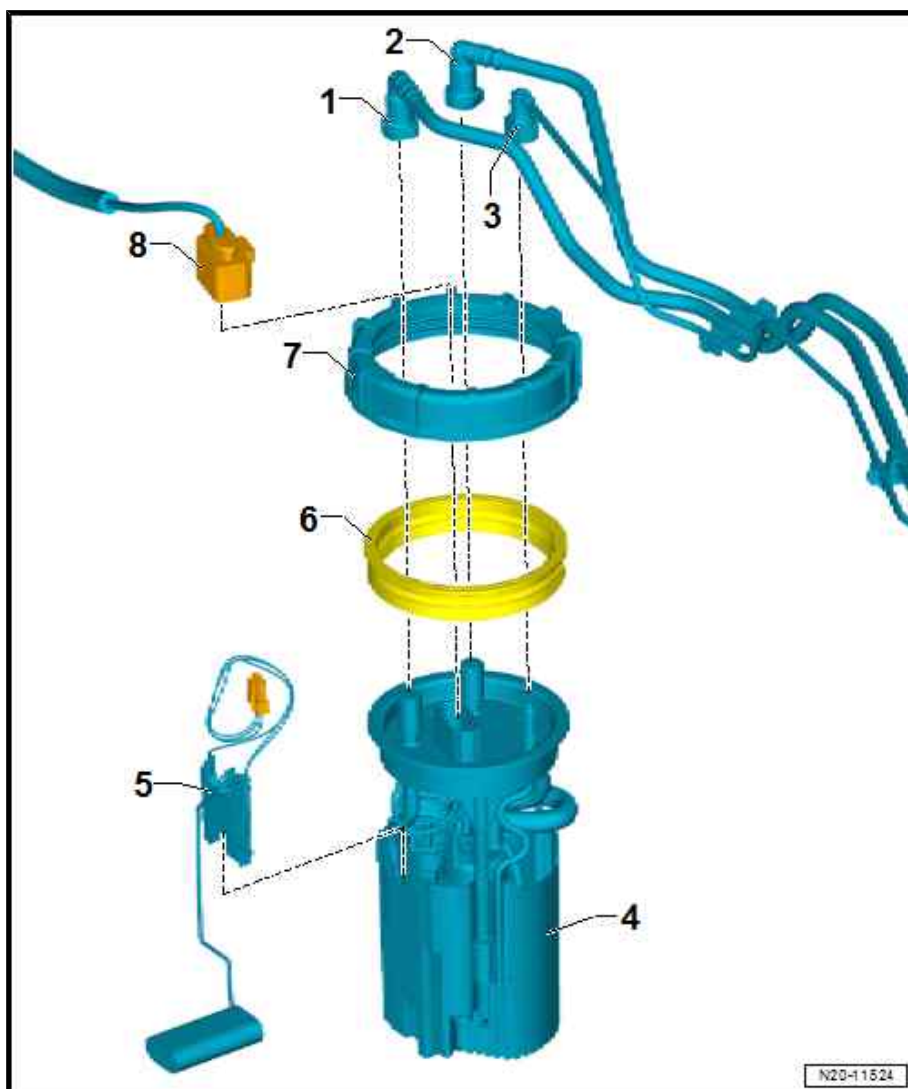
6 - Seal

- ☐ Renew after removal
- ☐ When installing, fit dry in fuel tank opening.
- ☐ Moisten with fuel only when installing fuel delivery unit - GX1-

7 - Union nut

- ☐ 110 Nm

8 - Electrical connector





3.2 Removing and installing fuel delivery unit, fuel gauge sender

Special tools and workshop equipment required

- ◆ Key for union nut - 3217-



- ◆ Torque wrench - V.A.G 1332-



Removing

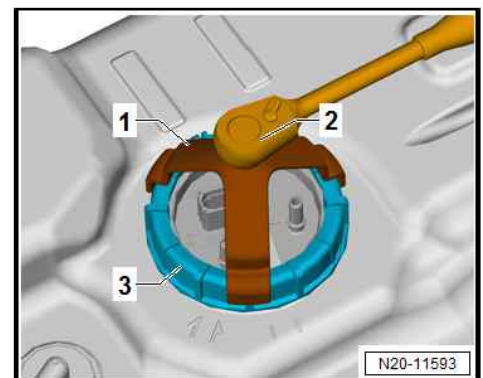
- Remove fuel tank ⇒ [page 19](#) .
- Unscrew union nut -3- with union nut tool - 3217- -1- and ratchet wrench -2-.



Note

When removing the fuel delivery unit - GX1- , make sure that the float arm of the fuel gauge sender - G- is not bent.

- Carefully pull out fuel delivery unit - GX1- with fuel gauge sender - G- through opening of fuel tank.
- Turn and tilt it as necessary while doing so.
- Place a cloth underneath the fuel delivery unit - GX1- to catch any escaping fuel.





Installing

Install in reverse order of removal, observing the following:

- Insert seal for fuel delivery unit - GX1- dry into opening of fuel tank.
- Coat inner side of seal with fuel.



Note

- ◆ *When inserting the fuel delivery unit - GX1- , ensure that the fuel gauge sender - G- is not bent.*
- ◆ *When tightening the union nut, be careful not to damage or crush the seal.*

Mark on flange -1- must align with mark on fuel tank -arrow-.

Torque settings

- ◆ ⇒ ["3.1 Assembly overview - fuel delivery unit and fuel gauge sender", page 26](#)

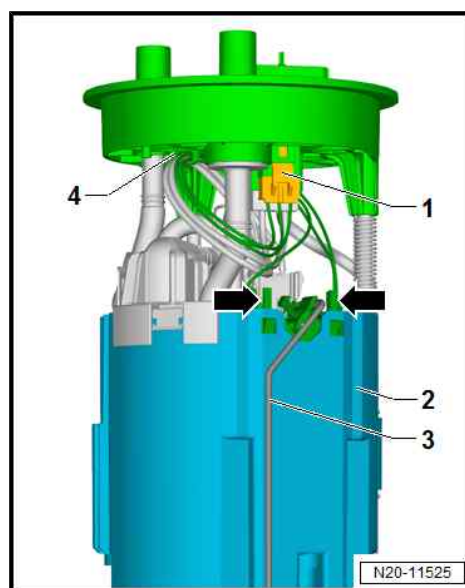
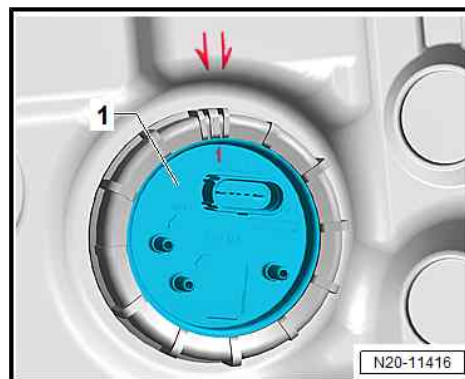
3.3 Removing and installing fuel gauge sender - G-

Removing

- Remove fuel delivery unit - GX1- ⇒ [page 27](#) .
- Remove connector -1- for fuel gauge sender - G- -3-.
- Unclip electrical wires from retainer -4-.
- Release retaining tab -arrows- by pulling off fuel gauge sender - G- -3- upwards from fuel delivery unit - GX1- -2-.

Installing

Install in reverse order of removal.





4 Plug-in connectors

⇒ ["4.1 Separating plug-in connectors", page 29](#)

4.1 Separating plug-in connectors

Assignment of plug-in connectors



Note

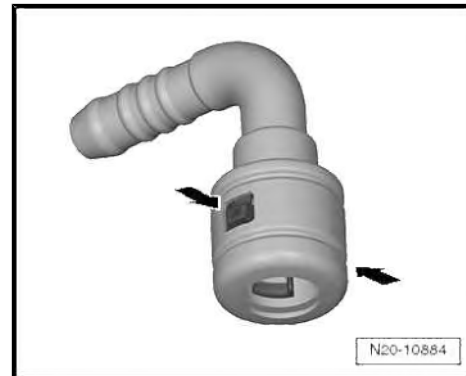
- ◆ *The plug-in connectors for fuel, vacuum and breather lines are colour coded. There is either a coloured dot on the plug-in connector or the release button is of the respective colour
 ⇒ [page 29](#).*
- ◆ *Plug-in connectors for fuel lines must engage »audibly« when joined.*
- ◆ *Ensure plug-in connector is secured properly by pulling it.*

Colour assignment

Plug-in connector	Colour code on plug-in connector
Fuel supply	black
Fuel return line	blue
Breather	White, beige
Vacuum	green

Version 1

Plug-in connector with release buttons -arrows- on right and left:

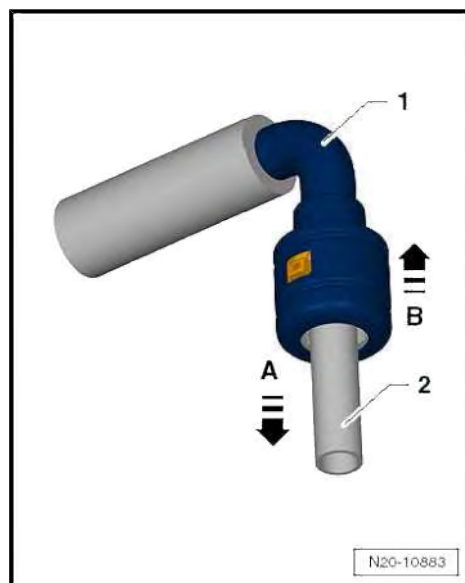




Open

- Push plug-in connector -1- in -direction of arrow A-.
- Press and hold release buttons.
- Pull plug-in connector -1- off fuel line -2- in -direction of arrow B-.

Note colour assignment during installation ➔ [page 29](#) .



Version 2

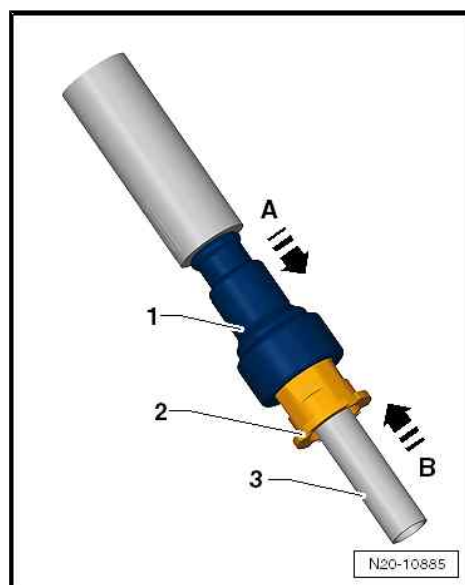
Plug-in connector with pull-release mechanism -arrow-.



Open

- Push plug-in connector -1- in -direction of arrow A-.
- Pull pull-release mechanism -2- in -direction of arrow B-.
- Pull plug-in connector -1- off fuel line -3- in -direction of arrow B-.

Note colour assignment during installation ➔ [page 29](#) .





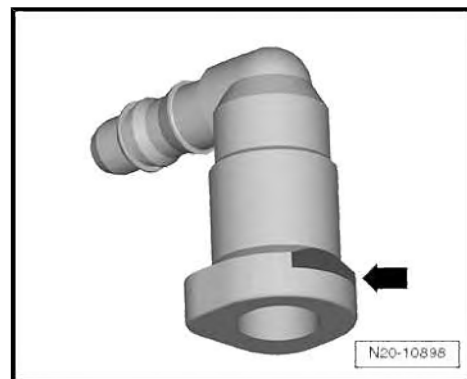
Version 3

Plug-in connector with button in front -arrow-

Open

- Press release button -arrow- and pull off plug-in connectors.

Note colour assignment during installation ⇒ [page 29](#) .



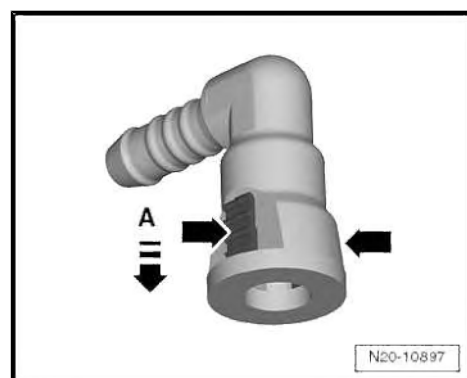
Version 4

Plug-in connector with release button -arrows- on right and left:

Open

- Push plug-in connector in -direction of arrow A-.
- Press release buttons -arrows- and pull off plug-in connector.

Note colour assignment during installation ⇒ [page 29](#) .



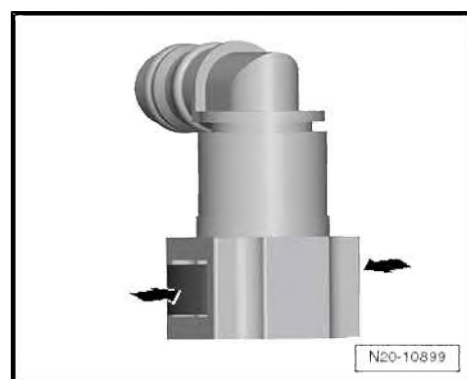
Version 5

Plug-in connector with release buttons -arrows- on right and left:

Open

- Press release buttons -arrows- and pull off plug-in connector.

Note colour assignment during installation ⇒ [page 29](#) .



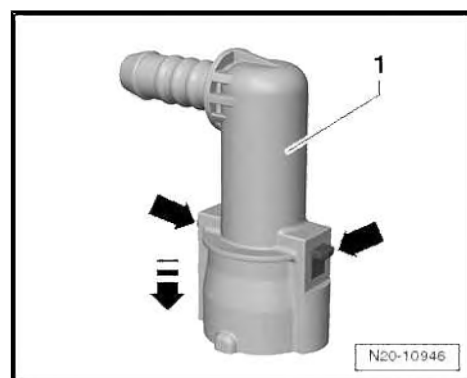
Version 6

Plug-in connector with release buttons -arrows- on right and left:

Open

- Push plug-in connector -1- in -direction of arrow-, and hold it in that position.
- Press release buttons -arrows- and pull off plug-in connector.

Note colour assignment during installation ⇒ [page 29](#) .





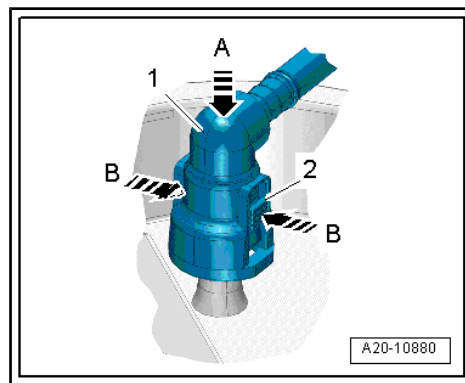
Version 7

Plug-in connector -1- with release buttons -2- on right and left:

Open

- Push plug-in connector -1- in -direction of arrow A-, and hold it in that position.
- Press release button -2- in -direction of arrow B- and pull off plug-in connector -1-.

Note colour assignment during installation ⇒ [page 29](#) .





5 Fuel filter

⇒ ["5.1 Assembly overview - fuel filter", page 33](#)

⇒ ["5.2 Removing and installing fuel filter", page 34](#)

5.1 Assembly overview - fuel filter

1 - Fuel filter

- ☐ Removing and installing
⇒ [page 34](#)

2 - Fuel filter support

- ☐ On some equipment
versions with crash protection
- ☐ Removing and installing
⇒ [page 36](#)

3 - Bolts

- ☐ Qty. 2
- ☐ 12 Nm

4 - Bolt

- ☐ 8 Nm

5 - Clip

6 - Bleeder valve

- ☐ Depending on equipment
- ☐ 8 Nm

7 - Seal

- ☐ Depending on equipment
- ☐ Renew after removal

8 - Water separator sender - G63-

- ☐ Depending on equipment
- ☐ Removing and installing
⇒ [page 46](#)

9 - Cover

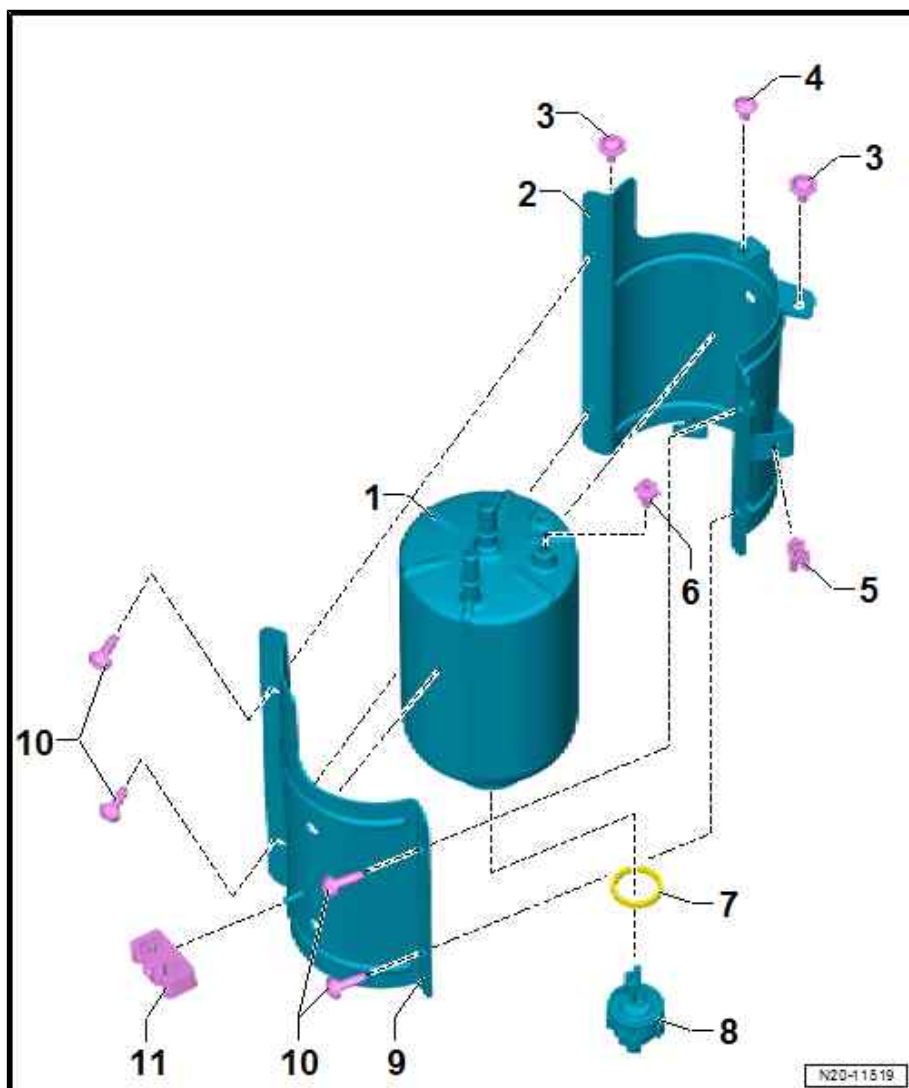
- ☐ On some equipment
versions with crash protection
- ☐ Removing and installing ⇒ [page 36](#)

10 - Bolts

- ☐ Qty. 4
- ☐ 8 Nm

11 - Bracket

- ☐ For fuel lines





5.2 Removing and installing fuel filter

⇒ [“5.2.1 Removing and installing fuel filter with water separator sender G63”, page 34](#)

⇒ [“5.2.2 Removing and installing fuel filter without water separator sender G63”, page 35](#)

⇒ [“5.2.3 Removing and installing bracket for fuel filter”, page 36](#)

5.2.1 Removing and installing fuel filter with water separator sender - G63-

Special tools and workshop equipment required

◆ Drip tray for workshop hoist - VAS 6208-



◆ Torque wrench - V.A.G 1783-



Removing

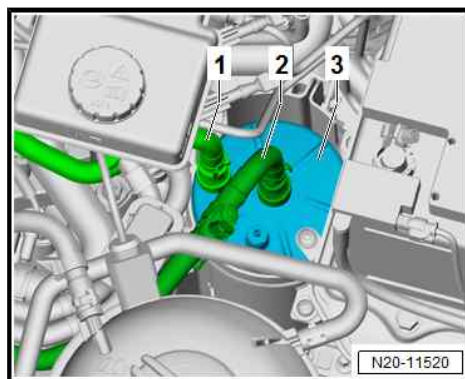
- Place drip tray for workshop hoist - VAS 6208- beneath fuel filter.

CAUTION

The fuel system is pressurised.

Danger of injury caused by fuel spray.

- Wear protective goggles.
- Wear protective gloves.
- To release pressure, wrap a clean cloth around the connection and carefully loosen the connection.



- Disconnect fuel line -1- on fuel filter -3-.
- Disconnect fuel line -2- on fuel filter -3-.



- Unclip electrical wire -3-.
- Unclip fuel lines -4-.
- Unscrew bolts -2-.

i Note

The water separator sender - G63- is screwed into the fuel filter from underneath. When laying the fuel filter aside, consider the length of the electrical wire of the water separator sender - G63- .

- Move fuel filter -1- with bracket clear to one side.
- Disconnect connector -2- of water separator sender - G63- at fuel filter -1-.
- Remove fuel filter -1- with bracket.

Change fuel filter

- Remove water separator sender - G63- [⇒ page 46](#) .
- Remove bracket for fuel filter [⇒ page 36](#) .

Installing

Install in reverse order of removal, observing the following:

- Renew seal for water separator sender - G63- .
- Bleed fuel system ⇒ Rep. gr. 23 ; Injection system; Filling/bleeding fuel system .

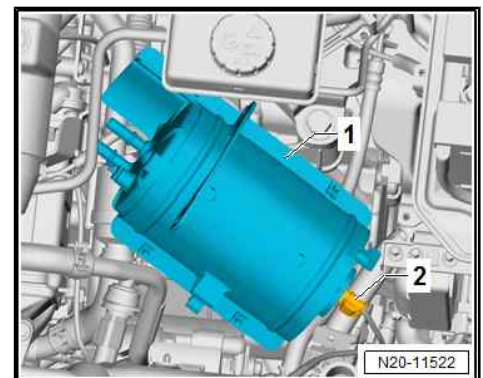
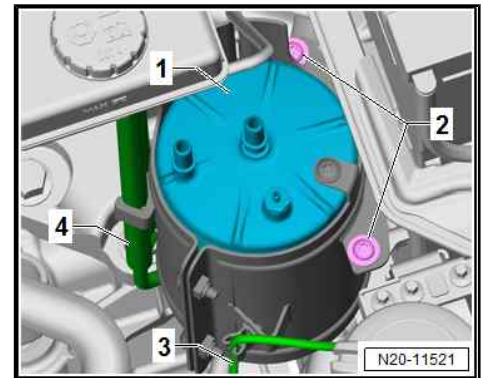
Torque settings

- ◆ [⇒ "5.1 Assembly overview - fuel filter", page 33](#)

5.2.2 Removing and installing fuel filter without water separator sender - G63-

Special tools and workshop equipment required

- ◆ Drip tray for workshop hoist - VAS 6208-





- ◆ Torque wrench - V.A.G 1783-



Removing

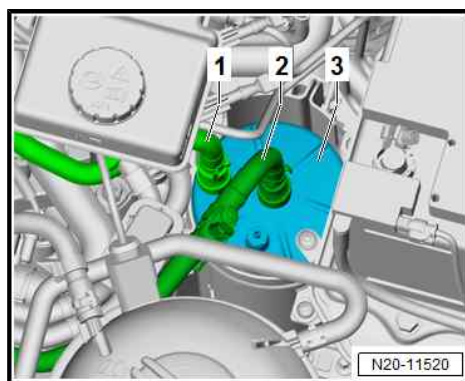
- Place drip tray for workshop hoist - VAS 6208- beneath fuel filter.

CAUTION

The fuel system is pressurised.

Danger of injury caused by fuel spray.

- Wear protective goggles.
- Wear protective gloves.
- To release pressure, wrap a clean cloth around the connection and carefully loosen the connection.



- Disconnect fuel line -1- on fuel filter -3-.
- Disconnect fuel line -2- on fuel filter -3-.
- Unclip electrical wire -3-.
- Unclip fuel lines -4-.
- Unscrew bolts -2-.

Change fuel filter

- Remove bracket for fuel filter ⇒ [page 36](#) .

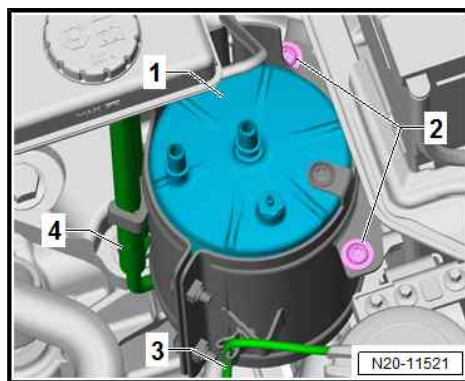
Installing

Install in reverse order of removal, observing the following:

- Bleed fuel system ⇒ Rep. gr. 23 ; Injection system; Filling/bleeding fuel system .

Torque settings

- ◆ ⇒ [“5.1 Assembly overview - fuel filter”, page 33](#)



5.2.3 Removing and installing bracket for fuel filter

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1783-



Removing

- Remove fuel filter ⇒ [page 35](#) .
- Unscrew bolts -arrows-, and remove bracket -4- and (if fitted) crash protection.
- Unscrew bolt -1-. Remove bracket -3- from fuel filter -2-.

Installing

Install in reverse order of removal, observing the following:

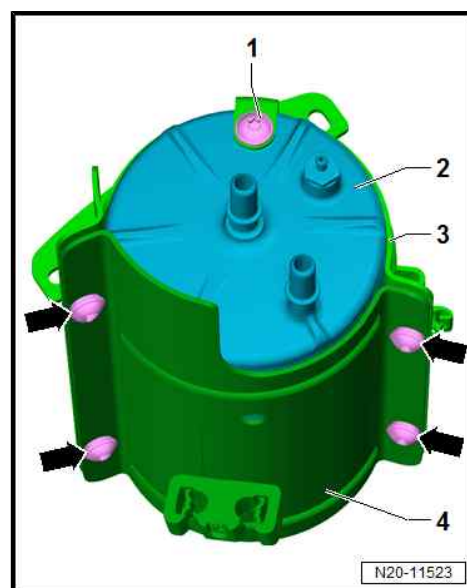


Note

If a new bracket for the fuel filter is installed, all fastening elements must be fitted on the bracket.

Torque settings

- ◆ ⇒ [“5.1 Assembly overview - fuel filter”, page 33](#)





6 Accelerator pedal

⇒ [“6.1 Assembly overview - accelerator module”, page 38](#)

⇒ [“6.2 Removing and installing accelerator pedal module GX2”, page 38](#)

6.1 Assembly overview - accelerator module

1 - Bolts

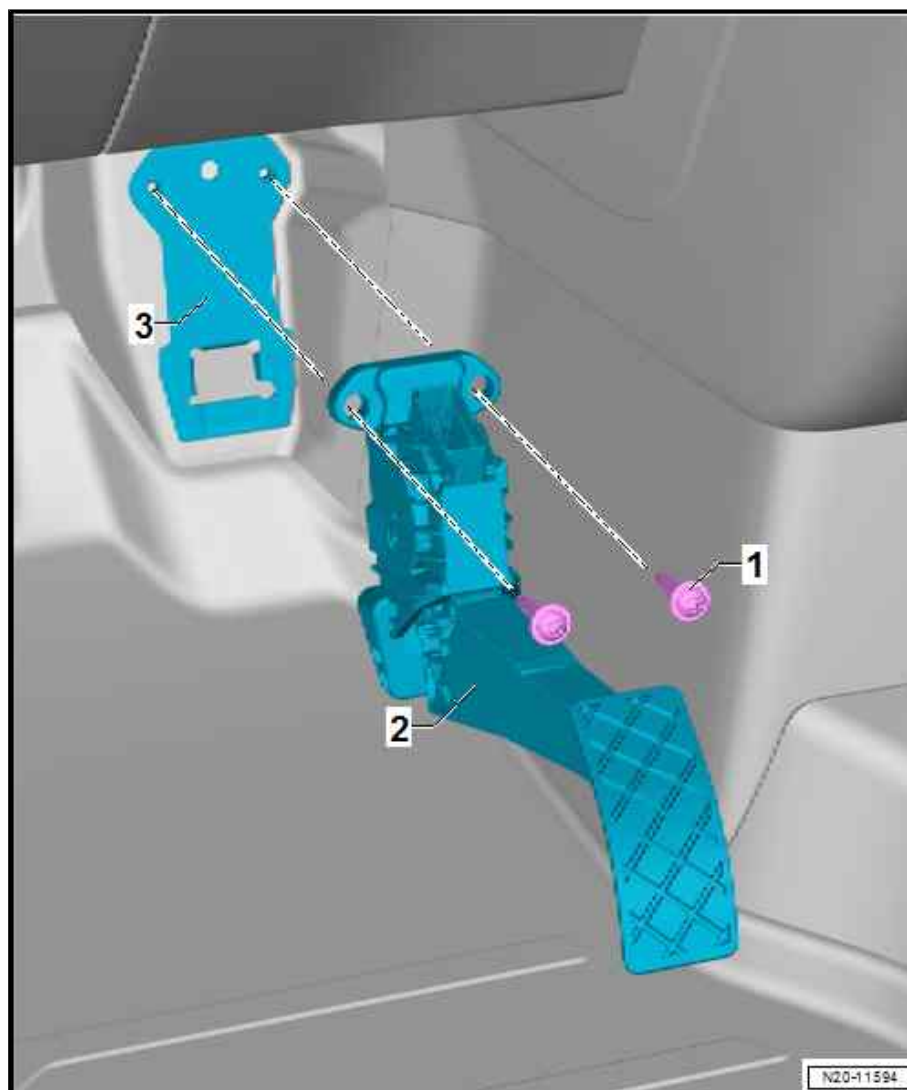
- ☐ Qty. 2
- ☐ 6 Nm

2 - Gas pedal module - GX2-

- ☐ Removing and installing
⇒ [page 38](#)

3 - Support

- ☐ for accelerator pedal
module - GX2-



6.2 Removing and installing accelerator pedal module - GX2-

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1783-



Removing

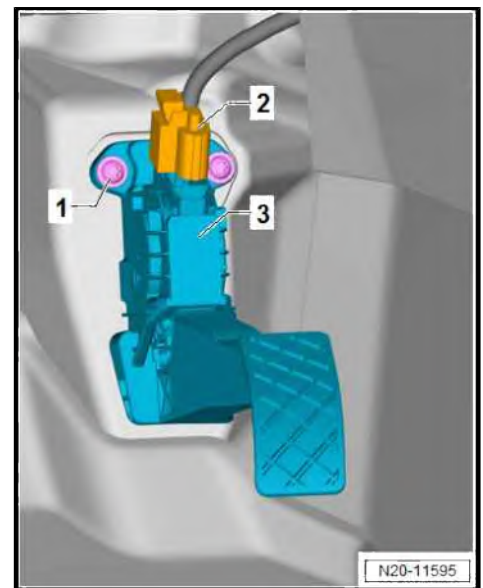
- Separate electrical connector -2-.
- Unscrew bolts -1-.
- Remove accelerator module - GX2- -3-.

Installing

Install in reverse order of removal, observing the following:

Torque settings

- ◆ ⇒ [“6.1 Assembly overview - accelerator module”, page 38](#)





7 Fuel pump

⇒ ["7.1 Checking fuel system pressurisation pump G6", page 40](#)

7.1 Checking fuel system pressurisation pump - G6-

⇒ ["7.1.1 Checking fuel pressure", page 40](#)

⇒ ["7.1.2 Checking fuel delivery rate", page 43](#)

7.1.1 Checking fuel pressure

Special tools and workshop equipment required

- ◆ Drip tray for workshop hoist - VAS 6208-



- ◆ Hose adapter set - VAS 6551/5-



- ◆ Pressure gauge - VAS 6551-
- ◆ Vehicle diagnostic tester

Prerequisites for check

- The event memory of the engine control unit contains no event entry relating to the voltage supply to the fuel pump.
- Battery charger connected ⇒ Electrical system; Rep. gr. 27 ; Battery; Charging battery .
- Fuse for fuel pump OK. Fuse assignment ⇒ Current flow diagrams, Electrical fault finding and Fitting locations
- Fuel filter OK.
- Visual check carried out
- Fuel tank at least $\frac{3}{4}$ full
- Ignition off



Test sequence

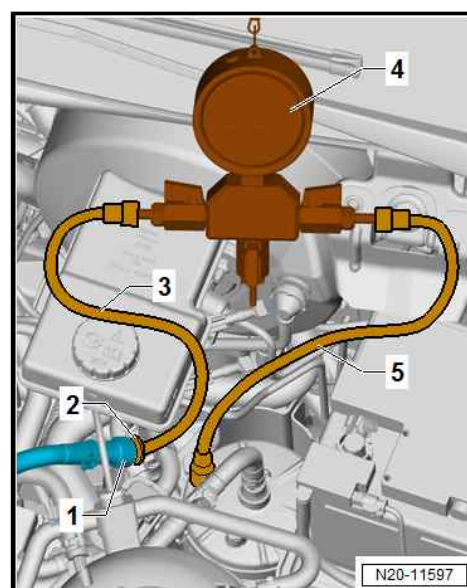
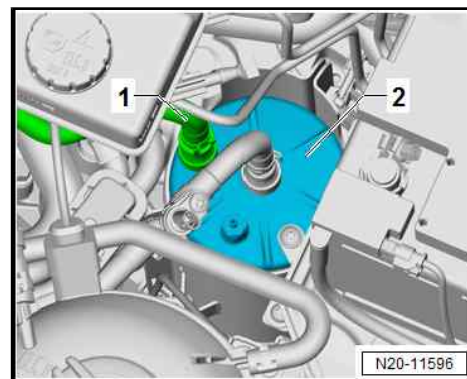
CAUTION

The fuel system is pressurised.

Danger of injury caused by fuel spray.

- Wear protective goggles.
- Wear protective gloves.
- To release pressure, wrap a clean cloth around the connection and carefully loosen the connection.

- Place drip tray - VAS 6208- beneath vehicle.
- Disconnect fuel line -1- from fuel filter -2-.
- Collect escaping fuel with a cleaning cloth.
- Connect pressure tester - VAS 6551- -4- to hose adapter - VAS 6551/5-4- -5- and fuel filter.
- Connect pressure tester - VAS 6551- -4- to hose adapter - VAS 6551/5-3- -3-.
- Fit connection of hose adapter - VAS 6551/5-3- -2- to fuel line -1-.





- Switch on pressure tester - VAS 6551- -1- by pressing **[On/Off]** button.
- Open shut-off taps -A- and -B- of pressure tester - VAS 6551- -1-.
- Close shut-off valve -C-.
- Actuate with fuel system pressurisation pump - G6- with ➔ Vehicle diagnostic tester.



Note

The fuel pump will be activated for 30 seconds.

- Read pressure shown by pressure tester - VAS 6551- . Specification is 4.5 bar at least.
- Start engine and allow to idle.
- Read pressure shown by pressure tester - VAS 6551- . Specification is 4.5 bar at least.
- Place pressure tester - VAS 6551- onto windscreen so that it can be read from inside vehicle.
- Increase idling speed to 2500 rpm.
- Read pressure shown by pressure tester - VAS 6551- . Specification is 4.5 bar at least.

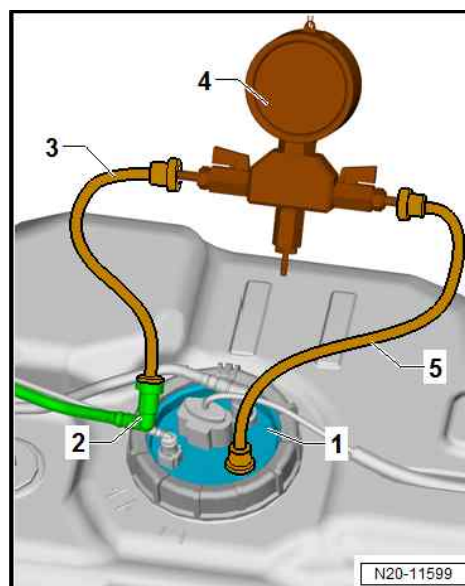
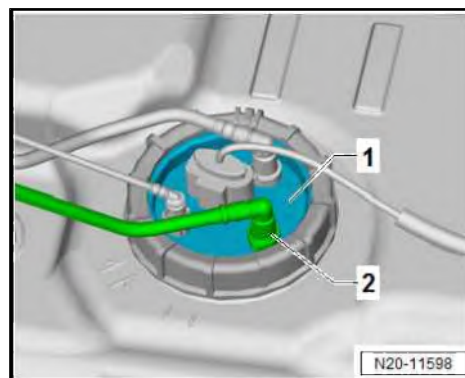
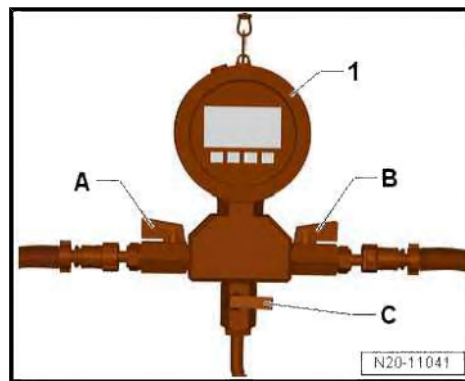
If pressure is not reached



Note

If the specified pressure is again not achieved, perform a measurement directly on fuel delivery unit - GX1- .

- Lower fuel tank ➔ [page 22](#) .
- Disconnect fuel line -2- from fuel delivery unit - GX1- -1-.
- Connect pressure tester - VAS 6551- -4- to hose adapter - VAS 6551/5-4- -5- and fuel delivery unit - GX1- -1-.
- Connect pressure tester - VAS 6551- -4- to hose adapter - VAS 6551/5-3- -3-.
- Connect hose adapter - VAS 6551/5-3- -3- to fuel line -2-.





- Switch on pressure tester - VAS 6551- -1- by pressing **[On/Off]** button.
- Open shut-off taps -A- and -B- of pressure tester - VAS 6551- -1-.
- Close shut-off valve -C-.
- Actuate with fuel system pressurisation pump - G6- with ➔ Vehicle diagnostic tester.



Note

The fuel pump will be activated for 30 seconds.

- Read pressure shown by pressure tester - VAS 6551- . Specification is 4.5 bar at least.
- Start engine and allow to idle.
- Read pressure shown by pressure tester - VAS 6551- . Specification is 4.5 bar at least.
- Place pressure tester - VAS 6551- onto windscreen so that it can be read from inside vehicle.
- Increase idling speed to 2500 rpm.
- Read pressure shown by pressure tester - VAS 6551- . Specification is 4.5 bar at least.

If specification is not attained

- Checking fuel delivery rate ➔ [page 43](#) .

If the specifications are still not attained:

- Renew fuel delivery unit - GX1- ➔ [page 27](#) .

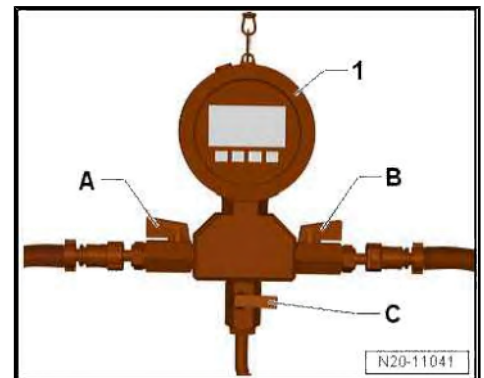
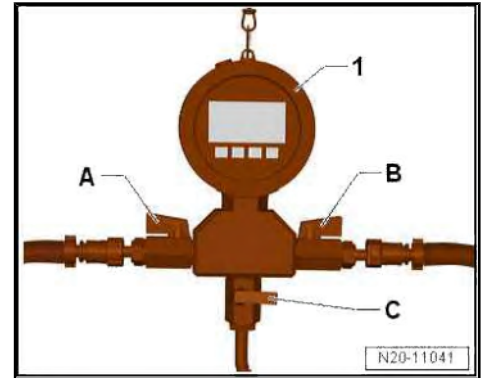
If specification is attained

- Remove pressure tester - VAS 6551- , and re-connect fuel supply line.



Note

If the pressure tester - VAS 6551- -1- is to be removed, open the shut-off tap -C-, and release remaining pressure. To do this, hold the hose of the shut-off tap -C- in a measuring container.



7.1.2 Checking fuel delivery rate

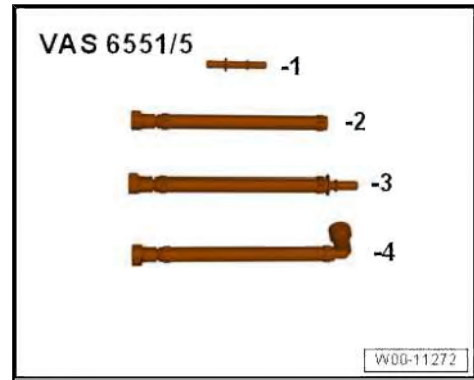
Special tools and workshop equipment required

- ◆ Drip tray for workshop hoist - VAS 6208-





- ◆ Hose adapter set - VAS 6551/5-



- ◆ Pressure gauge - VAS 6551-
- ◆ Vehicle diagnostic tester

Test conditions:

- Delivery pressure of fuel pump checked beforehand.
- Connect charger or battery voltage above 12.5 volts.
- Fuel tank at least $\frac{3}{4}$ full

Test sequence

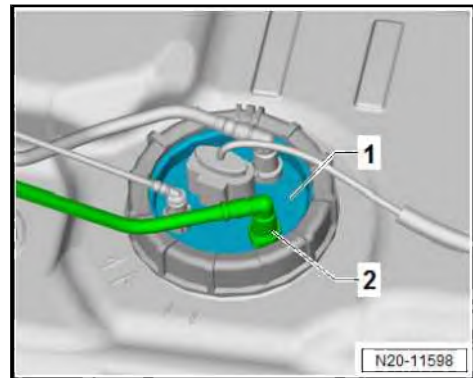
- Lower fuel tank ➔ [page 19](#) .

⚠ CAUTION

The fuel system is pressurised.

Danger of injury caused by fuel spray.

- Wear protective goggles.
- Wear protective gloves.
- To release pressure, wrap a clean cloth around the connection and carefully loosen the connection.

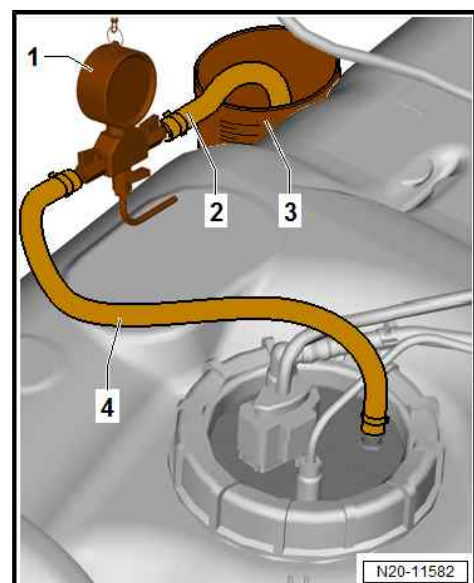


- Place drip tray - VAS 6208- beneath vehicle.
- Disconnect fuel supply line -2- from fuel delivery unit - GX1-1-.
- Collect escaping fuel with a cleaning cloth.
- Connect pressure tester - VAS 6551- -1- with connection cable - VAS 6551/5-4- -4- to fuel delivery unit - GX1-1- .
- Connect connection cable - VAS 6551/5-3- -2- to pressure tester - VAS 6551- .
- Extend connection cable - VAS 6551/5-3- -2- using hose - VAS 6551/2- , and hold end of hose - VAS 6551/2- in measuring beaker -3-.



Note

Ensure hose connections are fitted securely.





- Open shut-off taps -A- and -B- of pressure tester - VAS 6551- -1- in direction of flow.
- Close shut-off valve -C-.
- Switch on pressure tester - VAS 6551- by pressing **On/Off** button.
- Carry out “fuel pump activation”.
- Select “Check delivery rate”.

i Note

The fuel pump is now activated for 30 seconds.

- Slowly close shut-off tap -B- until back pressure of 4 bar is reached.
- Drain measuring container.
- Carry out guided function “Activating fuel pump” again.
- Select function “Check delivery rate”.

i Note

The fuel pump is now activated for 30 seconds.

Minimum flow rate according to specification: > 1.3 l/30 seconds.

If specification is not attained

- Perform the fuel delivery rate check one more time.

If the specifications are still not attained:

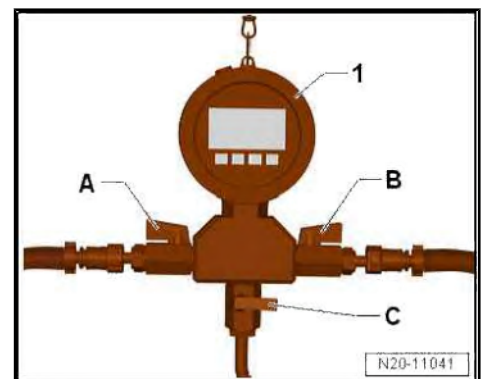
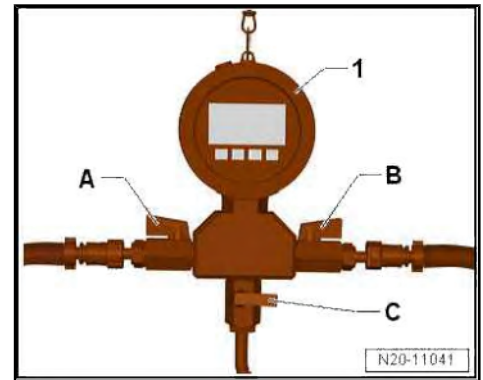
- Renew fuel delivery unit - GX1- ➔ [page 27](#) .

If specification is attained

- Remove pressure tester - VAS 6551- , and re-connect fuel supply line.

i Note

If the pressure tester - VAS 6551- -1- is to be removed, open the shut-off tap -C-, and release remaining pressure. To do this, hold the hose of the shut-off tap -C- in a measuring container.





8 Senders and sensors

⇒ ["8.1 Removing and installing water separator sender G63", page 46](#)

8.1 Removing and installing water separator sender - G63-

Special tools and workshop equipment required

- ◆ Drip tray for workshop hoist - VAS 6208-



Removing

- Remove fuel filter ⇒ [page 34](#) .



Note

Catch any escaping fuel with a drip tray - VAS 6208- or a cloth.

- Unscrew water separator sender - G63- -2- from fuel filter -1-.

Installing

Install in reverse order of removal, observing the following:

- Renew seal for water separator sender - G63- .

Torque settings

- ◆ ⇒ ["5.1 Assembly overview - fuel filter", page 33](#)

